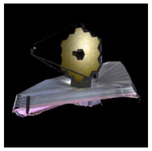

9903 - Preparing for the Worst: A ToO Program for Utilizing JWST for Critical Characterization of Potential Impactors

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
Observation Folder				
	2	SED Sequence	MIRI Imaging	(1) Potential-Impactor

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
	3	Lightcurve Sequence	NIRCam Imaging	(1) Potential-Impactor

ABSTRACT

Asteroid impacts have profoundly affected the evolution of life on Earth. Since the 1980s, the field of planetary defense has been developing plans and technology to prevent asteroid impacts if possible and mitigate their effects as necessary. If an object were on a collision course with the Earth, early knowledge of its characteristics would be vital to determine what to do next. JWST has unique capabilities to provide that early knowledge. We propose below a multi-cycle target of opportunity (ToO) program of MIRI and NIRCam observations to provide diameter uncertainty better than 25% and orbit improvements in case a potential impact threat to the Earth is identified, triggering upon release of a warning from the International Asteroid Warning Network. It is likely that JWST/MIRI will be the best, fastest way to obtain a useful size measurement for a potential impactor, as demonstrated with potential lunar impactor 2024 YR4. We do not expect that this specific program will be triggered, but having such a program in place is important, even if only to provide an easy framework for use in an emergency rather than only developing a program after an impact threat is identified.

OBSERVING DESCRIPTION

Asteroid impacts have profoundly affected the evolution of life on Earth. Over the last 30-40 years, the field of planetary defense has identified the scope of the threat and is working to develop plans and technology to prevent asteroid impacts if possible and mitigate their effects as necessary. In particular, if an object were on a collision course with the Earth, early knowledge of key impactor characteristics is vital to inform the next planetary defense actions [1,2]. JWST has unique capabilities to provide early knowledge of the potential threat, as shown by its contributions to determining the size and improving the orbit of the potential impactor 2024 YR4 in early 2025. We propose below a multi-cycle, zero-proprietary period, target of opportunity program of spectrophotometric, astrometric, and lightcurve measurements with MIRI and NIRCam to provide diameter uncertainty better than 25% and positional measurements unobtainable from the ground in case a potential impact threat to the Earth is found. If an impact threat to Earth is found, it is likely that JWST/MIRI will be the best, fastest way to obtain a useful size measurement for such an object. We do not expect that this program will be triggered during the three cycles we propose for it, but nevertheless we argue below for the utility of having such a program in place, even if only to provide an easy framework for use in case of emergency rather than starting a program after an impact threat is identified.

Generic Targets

#	Name	Criteria	Description
(1)	Potential-Impactor	Asteroid with H < 26 Found with > 1% Chance of Earth Impact in Next 50 Years	

Observation

Proposal 9903, Observation 2: SED Sequence

Fri Mar 13 18:01:32 GMT 2026

Diagnostic Status: Warning

Observing Template: MIRI Imaging

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Generic Targets

#	Name	Criteria	Description
(1)	Potential-Impactor	Asteroid with H < 26 Found with > 1% Chance of Earth Impact in Next 50 Years	

Template

Subarray
FULL

Dithers

#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size
1	CYCLING	1	4						DEFAULT

Spectral Elements

#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F1280W	FASTR1	27	4	1	Dither 1	4	16	1232.118	
2	F1000W	FASTR1	27	4	1	Dither 1	4	16	1232.118	
3	F1280W	FASTR1	27	4	1	Dither 1	4	16	1232.118	
4	F1500W	FASTR1	27	4	1	Dither 1	4	16	1232.118	
5	F1280W	FASTR1	27	4	1	Dither 1	4	16	1232.118	
6	F1000W	FASTR1	27	4	1	Dither 1	4	16	1232.118	
7	F1280W	FASTR1	27	4	1	Dither 1	4	16	1232.118	
8	F1500W	FASTR1	27	4	1	Dither 1	4	16	1232.118	
9	F1280W	FASTR1	27	4	1	Dither 1	4	16	1232.118	

Special Requirements

Target Of Opportunity Response Time 14 Days, Carry-Over

Observation

Proposal 9903, Observation 3: Lightcurve Sequence

Fri Mar 13 18:01:32 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Generic Targets

#	Name	Criteria	Description
(1)	Potential-Impactor	Asteroid with H < 26 Found with > 1% Chance of Earth Impact in Next 50 Years	

Template

Module	Subarray
B	FULL

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	INTRAMODULEBOX	4	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F150W2	F322W2	MEDIUM2	5	6	24	4	11037.4	
2	F150W2	F322W2	MEDIUM2	5	6	24	4	11037.4	

Special Requirements

Offset 40.0 arcsec, 40.0 arcsec
Target Of Opportunity Response Time 14 Days, Carry-Over